

Scope of Work - Replacement of (5) Water Source Heat Pumps at Glenmont Mansion

Location: Glenmont Mansion – Thomas Edison National Historical Park, 119 Park Way, West Orange, NJ 07052.

1. Background

The Glenmont Mansion, part of the Thomas Edison National Historical Park, utilizes a geothermal system to provide heating and cooling. (5) water source heat pumps, each rated at 5 tons, have reached the end of their service life and require replacement to maintain system efficiency and reliability.

2. Objectives

The objective of this project is to remove and replace (5) existing water source heat pumps with new, medium-efficiency units compatible with the existing geothermal loop system. The work must be completed in compliance with National Park Service standards, local building codes, and environmental regulations.

3. Scope of Work

A. Removal

- Disconnect electrical, piping, and controls from existing heat pumps.
- Drain and cap water lines as necessary.
- Remove existing heat pump units and associated supports.
- Dispose of old equipment per EPA and local regulations.

B. Installation

- Furnish and install five (5) new water source heat pumps, each rated at 5 tons.
- Ensure compatibility with the existing geothermal system.
- Reconnect piping, electrical, and controls.
- Install vibration isolation pads and insulate piping.
- Verify leak-free connections and proper mounting.

C. Testing and Commissioning

- Test heating and cooling operation for each unit.
 - Integrate into basic control capability to the Building automation system, KMC system
 - Balance water flow and verify system performance.
 - Provide start-up report and commissioning documentation.
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4. Technical Specifications

- Capacity: 5 tons per unit.
 - Efficiency: Minimum EER 15.0 and COP 4.5 or higher.
 - Refrigerant: EPA-approved
 - Controls: Compatible with existing building automation system.
 - Construction: Heavy-gauge galvanized steel cabinet with sound insulation.
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5. Warranty

- Equipment: Minimum 5 years on compressor, 1 year on parts.
 - Labor: Minimum 1 year on installation workmanship.
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6. Environmental & Sustainability Compliance

- Meet ASHRAE 90.1 energy efficiency standards.
 - Follow EPA refrigerant handling and disposal guidelines.
 - Minimize waste and recycle materials where possible.
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7. Deliverables

- Equipment submittals prior to installation.
- Operation and maintenance manuals.
- Training for NPS staff on system operation.

8. Period of Performance

- Duration: 90 calendar days from Notice to Proceed.
 - Work Hours: Monday through Friday, 7:30 AM to 3:30 PM.
 - No work on weekends or federal holidays.
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9. Safety and Compliance

- Comply with OSHA safety standards.
- Protect historic finishes and structures during work.
- Maintain clean and secure work areas.

Contracting Officer's Representative (COR): Peter Hepp

Project Manager: Tim Stauffer